# Idea generation techniques

Level 6 · Thinking tools · Worksheet

CREATIVE THINKING WITH MR KELLY

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ use mind maps, brainstorming and Crazy 8s
- ☐ use SCAMPER, random words and worst possible idea
- ☐ think laterally and borrow ideas from nature or other industries

## I DO — Worked example

Read how a full-mark answer is built

*Brief: reduce the amount of food wasted in your school canteen.*

**1 Use SCAMPER to generate three different ideas. Name the letter used for each.** (3 marks)

**Full-mark answer** each ✓ is one mark

Substitute: smaller plates so pupils take less food ✓. Eliminate: remove trays so people only carry what they'll eat ✓. Put to another use: leftover bread goes to a community fridge or is made into croutons ✓.

**Why marks are not awarded**

*"Smaller plates, smaller plates in a different colour, smaller bowls."*

1 point: ideas are all variations of one idea and no SCAMPER letters are named.

## WE DO — Guided practice

Use the starters and word bank to help you

*You want ideas for a reusable water bottle that encourages pupils to drink more.*

**2 Explain how biomimicry could inspire ideas.** (2 marks)

**Sentence starters**

One reason is … because …

Another reason is … so …

**Word bank**

solves · Biomimicry · problems · linked · nature · Brief · cactus · copying

Tip: 2 marks = 2 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** Which technique asks you to sketch eight ideas in eight minutes?
☐ Mind map ☐ SCAMPER ☐ Crazy 8s ☐ Six hats

**b)** In SCAMPER, what does the E stand for?
☐ Evaluate ☐ Eliminate ☐ Explore ☐ Expand

**c)** Velcro was inspired by burrs sticking to a dog. This is an example of...
☐ Reverse brainstorming ☐ Converging ☐ A survey ☐ Biomimicry

**d)** What is the first rule of brainstorming?
☐ No judging ☐ Only good ideas ☐ Work alone ☐ Pick one quickly

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Crazy 8s** | ____ | **A** Eight different ideas sketched in eight minutes |
| 2. **SCAMPER** | ____ | **B** How could we make it worse? Then reverse the answers |
| 3. **Worst possible idea** | ____ | **C** Force a link between a random word and the problem |
| 4. **Reverse brainstorm** | ____ | **D** Generate terrible ideas, then flip them into good ones |
| 5. **Random word technique** | ____ | **E** Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse |
| 6. **Morphological chart** | ____ | **F** Combine one option for each feature to make new ideas |

*Your group has run out of ideas for a library redesign.*

### 5 Describe how you would run a "worst possible idea" session. (3 marks)

### 6 Explain what is meant by ‘Lateral thinking’. (2 marks)

**7 Explain what is meant by ‘Biomimicry’.** (2 marks)

**Check and reflect**
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?